positive half wave and the other for the negative half wave.

For example, to generate a 60Hz waveform, each half cycle is 8ms long. In this time, you have to produce a pulse sequence. If you use 16 pulses for one half cycle (actually, we use much more), the time for each pulse is 0.5ms. In this time, the duty cycle or pulse width will vary from 0.1ms to 0.45ms in accordance with the sine wave shape. Such a waveform is shown in Fig. 2. It shows the PWM pin output waveform from PIC16F73 microcontroller for generating one half cycle of a sine waveform of about 60Hz frequency. The pulses are so close that their widths cannot be seen individually. The sine wave peak is brighter, as the width is greater in the middle.

The filtered current output would be sine shaped for each half cycle. If you change the time of each pulse to 0.625ms, the same 16 pulses will take a time of 10ms per half cycle, generating a 50Hz sine waveform.

Just 16 pulses will give sufficiently good wave shapes. To change the period of the sine waveform, use the sine look-up table, which shows 16 points per half cycle. The values are:

$\mathrm{Sin}(n \times \pi)/16$

For n = 1 to 16. The scaling is done by multiplying these values by 128. These are the look-up table entries.

If you take the table value and output it as a width on PWM pin, 16 such points will take only a few milliseconds. In order to vary the frequency, each of the table values is output as pulse width more than once. The number of times each table value is repeated determines the period.

For period variation, a potentiometer is used to apply a variable voltage to the analogue digital converter of the microcontroller. The converted value ranges from 1 to 255. It is divided by 16 to give values 1 to 16. Subtracting these values from number 25 gives 24 to 9 for the potmeter range. Each of the table points is repeated 24 times for the slowest speed.

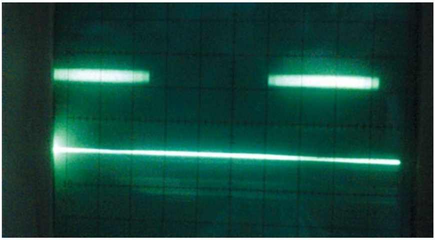

Fig. 2: The PWM pin output waveform from a DSP embedded controller for generating one half cycle of a sine waveform of about 60Hz frequency. The pulses are so close that their widths cannot be seen individually. The sine wave peak is brighter, as the width is greater in the middle

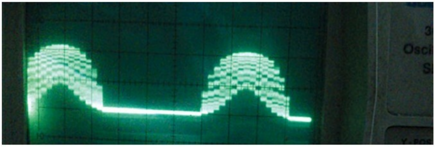

Fig. 3(a): The waveform of sine frequency PWM after a low-pass filter. The waveform smoothens when current passes through the fan winding inductance. The second PWM waveform will fill the other half cycles similarly

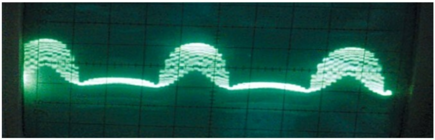

Fig. 3(b): The same waveform after frequency is increased to 65Hz. The above waves are taken on two PWM pins. The wave in the negative half cycle positions corresponds to the other PWM output pin

For the fastest speed, each table point is repeated only nine times. Nine times 16 points give one half cycle. So, if you take the fastest speed to be 66Hz, it is a period of 15/2ms for one half cycle. Thus, the time of picking each table value and outputting it on the PWM pin as width-modulated pulse becomes 7.5/(9x16)ms or about 52 microseconds. Timer 2 interrupt of PIC16F73 microcontroller is initialised to operate once every 52 microseconds.

Thus, each timer interrupt picks the table entry and loads the period register of the PWM. That generates a pulse on pins 12 and 13 of PIC16F73. The filtered output appears as shown in Fig. 3.

The program is written in Assembly language, and the hex file is created using MPLAB integrated development environment (IDE) or Oshon PIC simulator IDE. The configuration byte is 52H for the 20MHz crystal used in the PIC circuit. The